

DHRUV JYOTI DAS

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EDUCATION

SRM UNIVERSITY OF SCIENCE AND TECHNOLOGY | B. TECH CSE | CGPA: 9.0 | EXPECTED: 2027
IIT MADRAS | BS IN DATA SCIENCE | EXPECTED: 2027

TECHNICAL SKILLS

Programming: Java, C, Python, JavaScript

Web Development: Node.js, Express.js, React Native, HTML5, CSS3, WebSocket, REST API

Databases: MySQL, PostgreSQL, MongoDB

Data Science & ML: Pandas, NumPy, Scikit-learn, Vision Transformers (ViT), Matplotlib, Seaborn, Gabor Filtering

AI & LLMs: RAG pipelines, multi-turn CQG, DeepSeek, local LLMs, prompt structuring, WebRTC

Tools: Firebase, Render, Serveo, ngrok, Git/GitHub, Tableau

INTERNSHIP EXPERIENCE

SAMSUNG R&D RESEARCH INTERN

Mar 2025 – Jan 2026

- Designed a **deterministic ambiguity generation framework** over story-level knowledge graphs constructed from **1,000 CoQA narratives**; Formalized ambiguity as **relation reachability from multiple graph components or parent nodes**, enabling single-turn ambiguous and unambiguous question synthesis.
- Developed **multi-parent and single-parent relation discovery algorithms** using graph traversal and component analysis; Generated a **balanced ambiguity–non-ambiguity dataset** grounded in graph topology rather than surface linguistic heuristics.
- Proposed a **component-aware clarification algorithm** that iteratively narrows ambiguity by tracking reachability of target relations from candidate nodes; Modeled clarification as a sequence of constrained graph navigation steps with guaranteed convergence.
- Implemented a **can-contain-relation reachability map** to prune invalid clarification paths; Reduced clarification depth and eliminated spurious ambiguity by enforcing structural consistency across graph components.
- Released the dataset and generation pipeline for **Hugging Face distribution**, enabling reproducible research on ambiguity detection, clarification strategies, and graph-grounded conversational QA.

REAL-TIME LLM-POWERED CODE VULNERABILITY DETECTION

08/2025 – Present

(Research Internship, UROP, SRMIST)

- Proposed a recall-optimized framework** based on the **CodeBERT** architecture to identify C/C++ vulnerabilities; Leveraged a **30x weighted cross-entropy loss** and balanced under-sampling to address the severe class imbalance in the BigVul dataset.
- Designed an Explainable AI (XAI) validation suite** using attention heatmaps and **LIME** to interpret model decision-making; Revealed the model's priority for semantic control-flow boundaries and structural syntax over superficial variable naming patterns.
- Conducted a nine-step adversarial benchmarking suite**, including variable renaming and logic trap analysis, to evaluate model robustness; Demonstrated a negligible **0.02% confidence shift** under adversarial naming perturbations, proving high semantic invariance.
- Achieved State-of-the-Art (SOTA) performance** with a **1.00 Recall** and **0.99 F1-score**, significantly outperforming traditional GNN-based baselines like Devign and VulDeePecker.
- Optimized inference latency** to 122ms–445ms per function on T4 GPUs, validating the system's feasibility for real-time integration into enterprise **CI/CD pipelines**.

IMPROVING SPATIOTEMPORAL COLLABORATIVE LEARNING NETWORKS (STCLN) FOR CROP MAPPING

(Research Internship, IIT Indore)

08/2025 – Present

- Extended the **STCLN (Spatiotemporal Contrastive Learning Network)** framework for masked self-supervised learning on satellite image time series, targeting **label-scarce crop mapping** and heterogeneous agricultural landscapes.
- Redesigned components of the **spatial encoder** by incorporating **multi-scale convolutional tokenizers**, **relative positional embeddings**, and **patch-wise spectral standardization** to improve **spatial feature disentanglement** in complex agricultural parcels.
- Enhanced the **masked reconstruction decoder** with **temporal-aware cross-attention**, **spectral-channel gating**, and **pixel-level reconstruction heads** to better recover spatiotemporal reflectance patterns, reduce **temporal interpolation error**, and stabilize **latent representation learning**.
- Investigated encoder-decoder interactions with varying **spatiotemporal masking ratios**, **block masking schedules**, and **temporal stride policies**, and quantified downstream effects on **crop-type classification accuracy**, **F1 macro scores**, and **representation separability**.
- Conducted controlled experiments on **Sentinel-2-based crop time-series datasets** (internal + benchmark) evaluating **latent representation quality**, **masked reconstruction MSE/MAE**, and **fine-tuning performance** under **1–10% labelled regimes**.
- Logged training runs using **Weighs & Biases** and applied **CCA (Canonical Correlation Analysis)** and **t-SNE embedding visualization** to compare learned representations across architectural variants.
- Observed preliminary gains including **+8–12% F1 macro improvement** under low-label settings and **reduced temporal drift** during masked reconstruction, demonstrating the viability of **decoder-aware self-supervision** for remote sensing.

RESEARCH PROJECTS

XALIGN: HINDI-FOCUSED LLM ALIGNMENT PIPELINE

(Independent Research, 2025)

- Developed a **Hindi-focused alignment pipeline** to enhance **instruction-following**, **safety refusal**, and **context retention** in **open-weight transformer models**, addressing cross-lingual degradation and training data sparsity.
- Assembled a **14K-sample Hindi instruction dataset** combining **synthetic queries**, **human-curated prompts**, and **safety-critical preference pairs**; integrated **toxicity**, **bias**, and **hallucination** annotations using a semi-supervised labeling protocol.
- Implemented **Supervised Fine-Tuning (SFT)** with **QLoRA adapters** on **7B/13B parameter models**, followed by **Direct Preference Optimization (DPO)** using binarized human feedback to approximate a reward policy without RL rollouts.
- Trained a lightweight **reward model** for Hindi queries using contrastive preference data and calibrated it with **Expected Policy Divergence (EPD)** analysis to control over-optimization and mode collapse.
- Built a custom **evaluation harness** including **toxicity metrics (Perspective API proxy)**, **MT-Bench-style preference scoring**, **BLEU/ChrF++ for translation drift**, and **SafetyBench-HI**, a small internal benchmark to measure **refusal correctness** and **speculative compliance** in Hindi.
- Benchmarked aligned models against multilingual baselines (e.g., **Mistral-Instruct**, **Qwen2-7B-Instruct**) and observed **+27% improvement in preference score**, **–38% unsafe compliance**, and **reduced English fallback** during Hindi-only prompting.

TUMOR DETECTION USING NON-CNN IMAGE ANALYSIS

(Independent Research, 2025)

- Designed a **transformer-based tumor classification pipeline** using **ViT** and **Swin Transformer** backbones on **CT scan** data, achieving **92.4% accuracy** and **0.94 AUC**, outperforming **ResNet** and **DenseNet** baselines under identical settings.
- Implemented **Gabor-filter feature enhancement** and **Albumentations**-based preprocessing to boost **texture sensitivity** on **low-contrast tumor regions**.
- Trained using **PyTorch** with **mixed precision (AMP)**, **cosine LR scheduling**, and **early stopping** to reduce overfitting and increase training efficiency.
- Enabled **variable-resolution inputs** (224×224–512×512) via **adaptive pooling** and **attention-based tokenization**, eliminating the need for **padding or resizing**.
- Evaluated using **stratified splits**, **cross-entropy loss**, **ROC-AUC**, and **confusion matrix** analysis; verified robustness across multiple **CT tumor subsets**.

TLS-MIMICKING TROJAN PROXY FOR CENSORSHIP-RESISTANT CAMPUS NETWORKING

(Self-Deployed Network Research / Systems Study)

- Deployed **Trojan over TLS/443** on an **Oracle Free Tier VPS (Telangana)** with **Let's Encrypt** certificates to create a stealth **HTTPS-mimicking proxy** that bypasses campus firewall DPI and SNI filtering.
- Configured **v2rayN (Windows)** as a local **SOCKS5 transport**, updated **Oracle ingress rules** and **Linux routing/NAT tables**, and verified stable outbound **TCP/HTTPS tunneling** across restricted networks.
- Achieved seamless access to blocked web services and SSH-over-TLS while evaluating protocol limitations (e.g., **UDP-blocked gaming traffic**, QUIC filtering) under real campus network constraints.

CHAT X – REAL-TIME CHAT & AI VIDEO APP

- Implemented **real-time messaging backend** with authenticated user sessions, **channel-based routing**, and persistent storage using **MongoDB**.
- Built **WebRTC signaling and media pipeline** with **ICE candidate gathering**, **SDP negotiation**, and NAT traversal via **STUN/TURN** infrastructure.
- Integrated **peer-to-peer video calling** with transport renegotiation and **session recovery** under variable network conditions.
- Added **transformer-based summarization** and **sentiment tagging**, exposing affective cues through lightweight visual markers.
- Designed **latency-aware client UI** with reactive message rendering, **ephemeral interaction states** (typing, reactions), and cross-device synchronization.

CERTIFICATIONS

ORACLE JAVA SE 11 DEVELOPER

OCI 2025 CERTIFIED DATA SCIENCE PROFESSIONAL

NPTEL JAVA & DBMS

FOUNDATIONAL LEVEL CLEARED IN DATA SCIENCE AND PROGRAMMING FROM IITM